Leakage-Free Benchmarking of ALL vs. AML Leukemia Subtype Classification Across Three Gene-Expression Cohorts

Interim Report: Literature Review, Data Sets and Exploratory Data Analysis

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***Abstract*—Acute lymphoblastic (ALL) and acute myeloid leukemia (AML) need different therapies, and the Golub et al. microarray data set (72 patients, 7,129 probes) is the classic benchmark for separating them from gene expression. Because the data are tiny and very high-dimensional ($p \gg n$), several published accuracies near 100% rest on validation designs that appear to leak information. This interim report lays the foundation for a leakage-free comparison of machine learning (ML) methods. We (i) review 38 references with a documented search protocol, (ii) select two independent open cohorts with the same clinical question—the multicentre MILE study (GSE13159, 1,292 ALL/AML samples, Affymetrix HG-U133 Plus 2.0) and a paediatric cohort of Andersson et al. (GSE7186, 121 samples, two-colour cDNA arrays)—and (iii) run a fully scripted exploratory analysis. The Golub data contain no missing values but 28.7% negative intensities, which motivates clipping and a $\log_2$ transform; after that, PC1 alone separates the classes with an AUC of 0.984. A shift between the training and independent sets is measurable. Gene-wise t statistics correlate across cohorts (Spearman $\rho = 0.56$ Golub–MILE, $\rho = 0.44$ Golub–Andersson), and dataset identity, not class, dominates the pooled variance until each cohort is standardised. These findings fix the design of the final study: in-fold feature selection, nested cross-validation and per-cohort normalisation.**

***Index Terms*—leukemia, gene expression, microarray, exploratory data analysis, batch effect, nested cross-validation, small-sample learning**

I. Introduction

Clinical problem. ALL arises from lymphoid and AML from myeloid progenitor cells. Both present with similar symptoms and blasts in bone marrow, yet induction chemotherapy, risk stratification and transplant decisions differ substantially, so an incorrect lineage assignment directly affects treatment. In routine practice the distinction relies on morphology, cytochemistry, immunophenotyping and cytogenetics; gene-expression profiling was proposed as a single, objective assay that reproduces these labels and can reveal sub-classes [1], [2]. Throughout this work the task is called *ALL vs. AML subtype classification*: both classes are leukemias, so the label "non-cancer" used in the course module for ALL is biologically incorrect and is not used here.

Video: https://youtu.be/VIDEO-LINK-HERE. Code: https://github.com/mrayenesay/leukemia-all-aml-ml. Every number in this report is generated by `scripts/run_eda.py` and imported as a LaTeX macro; see Section VI.

Microarray technology. A DNA microarray measures the abundance of thousands of mRNA transcripts in parallel by hybridising labelled cDNA/cRNA to immobilised probes. Affymetrix oligonucleotide arrays (used by Golub and MILE) report one intensity per probe set; the older MAS 4 "average difference" (perfect-match minus mismatch) can be negative and is accompanied by a Present/Marginal/Absent (P/M/A) detection call. Two-colour spotted cDNA arrays (Andersson) co-hybridise the sample (Cy3) with a common reference RNA (Cy5) and report a log ratio.

Research questions. The project tests four falsifiable questions:

- **RQ1** (method comparison): Under a leakage-free 5×5 nested stratified cross-validation (CV) on all 72 Golub samples, do penalised logistic regression, a support vector machine (SVM) and a random forest differ in balanced accuracy? H_0: equal mean balanced accuracy, tested pairwise with the corrected resampled t-test [3] and Holm adjustment.
- **RQ2** (protocol): Does the single 38/34 split of the course module yield an estimate inside the 95% confidence interval of nested CV, or is it optimistic or pessimistic?
- **RQ3** (generalisability): Is the ALL/AML signal shared across cohorts and platforms, i.e. is the correlation of gene-wise test statistics $\rho > 0$ and is the top-100 gene overlap larger than chance? In the final report: does a model trained on one cohort keep balanced accuracy ≥ 0.90 on another?
- **RQ4** (bonus): Do a deep model (MLP/autoencoder) and a tabular transformer (TabPFN) outperform the best classical model under the identical protocol?

RQ3 is answered in part already by the exploratory analysis in Section IV.

Contributions of this interim report. (1) a documented review of 38 references (Sec. II); (2) two harmonised external

cohorts, three data sets in total (Sec. III); (3) a scripted exploratory data analysis (EDA) with numbered modelling implications (Sec. IV); (4) a pre-specified plan for the final report (Sec. V).

II. Literature Review

Search protocol. We queried Crossref (keyword search filtered to journal articles), Europe PMC (full text) and PubMed/NCBI E-utilities on 3 Nov 2026. Discovery was restricted to 2021–2026; classic papers (1995–2020) were added by identifier and verified the same way. Queries were, e.g., *"golub leukemia gene expression classification"*, *"ALL AML leukemia microarray feature selection classification"*, *"machine learning cancer classification gene expression data review"* and the full-text query *"7129" AND "72 samples" AND (leukemia OR leukaemia) AND classification*. *Inclusion:* peer-reviewed journal or conference papers about the data sets used, classification or feature selection on expression data, validation bias, batch effects, or recent tabular learning. For papers with a Golub result we recorded the sample and gene counts they report. *Exclusion:* conference abstracts, image-based leukemia studies, papers whose Golub-labelled result could not be confirmed, and unindexed venues. Of 152 records screened by title, 24 abstracts were assessed; 30 studies and 8 method/statistics references were included. All 38 are peer-reviewed and 11 date from 2021 or later. DOIs were verified against Crossref; 31 works were consulted in full text, 6 at abstract level and 1 at citation level, and the claims below were checked against them with AI assistance (`reports/literature/claim_audit.md`).

Data sets and early studies. Golub et al. [1] introduced class discovery and class prediction with a 50-gene weighted-voting predictor learnt on 38 samples and applied to 34 independent samples. Armstrong et al. [4] showed that lymphoblastic leukemias with MLL translocations form a distinct expression class. MILE standardised protocols across 11 laboratories [5] and reached 92.2% cross-validated concordance with the gold standard over 18 classes and 88.1% in an independent prospective cohort [2]. Andersson et al. [6] classified 121 consecutive childhood leukemias with k-nearest neighbours. CuMiDa [7] curates such data sets for benchmarking.

Classifiers and gene selection. SVMs [8] were applied early to microarray classification [9], [10], and SVM-RFE introduced recursive backward elimination of genes [10]. Dudoit et al. [11] found that simple diagonal discriminant analysis and k-NN rival complex methods, Tibshirani et al. [12] proposed nearest shrunken centroids, and Díaz-Uriarte and Alvarez de Andrés [13] used random forests [14] for gene selection. Statnikov et al. [15] found multicategory SVMs strongest across 11 data sets, and the elastic net [16] offers sparse models that group correlated predictors. A review of 132 papers found hybrid feature selection the most frequent research direction (34.9%) [17]; another groups selection into filter, wrapper, embedded and hybrid methods [18].

Validation bias and batch effects. Selecting genes on all samples before CV produces optimistic error rates [19], and so does tuning hyper-parameters on the evaluation folds. Nested CV avoids this bias and gave almost unbiased, slightly pessimistic estimates in simulation [20]; the variance of the selection criterion remains a concern [21]. Tests on resampled estimates must account for overlapping training sets; the correction of [3] was derived for random subsampling, and rank tests are recommended across multiple data sets [22]. Batch effects can dominate expression data [23] and can be adjusted with an empirical Bayes location/scale method (ComBat) [24].

Recent work. AML can be separated from all other samples, including ALL, in 12,029 transcriptomes from 105 studies, also with study-disjoint splits [25], and swarm learning outperforms single-node models on simulated multi-site partitions of such data [26]. Recent results labelled as Golub are very high [27]–[30]. As far as the papers describe their protocols, however, [27], [30] select genes before the CV loop, [29] uses random hold-outs with selection and tuning on the same data (and lists CML, CLL and healthy classes that Golub does not contain), and [28] does not state whether selection is repeated in each fold. On medium-sized, low-dimensional tabular benchmarks, tree ensembles beat deep networks [31], a regime that excludes $p \gg n$ microarrays. FT-Transformer was the best deep model on most of 11 data sets without always beating boosted trees [32], TabPFN outperformed tuned tree ensembles on up to 10,000 samples and 500 features [33], and GenePT embeds gene text summaries with a language model [34].

Gap. In the four recent studies of Table I, accuracies lie between 97% and almost 100%, whereas earlier work with fixed or external validation reports lower figures (e.g. 2/34 test errors [12], about 5% error after correcting for selection bias [19]). The numbers are hard to compare: protocols mix LOOCV, k-fold CV, the fixed split and random hold-outs, some studies select genes on the full data [10], [27], [30], and none of the four recent studies reports a confidence interval. Most evaluate further data sets, but separately: in the Golub-based classification papers we read, no classifier or gene list is transferred to an independent leukemia cohort or platform. This project addresses these issues with one leakage-free nested protocol, interval estimates and the same pipeline on two external cohorts.

III. Data Sets

Table II summarises the three data sets; their class distributions are plotted in Appendix C (Fig. 7).

Golub (main data set). Golub et al. [1] (DOI 10.1126/science.286.5439.531), obtained from the Kaggle mirror of the original Broad Institute files ("Gene expression data set (Golub et al.)", licence CC0). Platform: Affymetrix Hu6800/HuGeneFL (GEO GPL80), 7,129 probe sets (58 AFFX controls; 6,434 map to a gene symbol), MAS 4 average difference with P/M/A calls. Most patients were children (all 47 ALL and 10 of the 25 AML cases) and 62 of the 72 samples are bone marrow; the independent set is more heterogeneous (peripheral blood, childhood AML, different reference laboratories) [1]. The reference split has 38 training

TABLE I
LITERATURE SUMMARY (SUBSET; THE FULL 30-STUDY TABLE IS IN APPENDIX A). LEAKAGE RISK IS OUR READING OF THE REPORTED PROTOCOL ORDER.

Ref. (year)	Data set	Pre-processing / feature selection	Method	Validation	Best result	Leakage risk
[1] (1999)	Golub 38/34	class-correlation (SNR) gene selection	50-gene weighted voting	LOOCV + fixed independent test	test: 29/34 called, all correct	low (selection on train)
[11] (2002)	Golub, NCI60, lymphoma	clip, log, univariate filter	DLDA, k-NN, trees	150 repeated 2:1 splits	DLDA median error 0	low
[19] (2002)	colon, Golub	selection inside vs. outside CV	SVM-RFE; Fisher + forward selection	external 10-fold, .632+	selection bias about 5%	– (studies leakage)
[13] (2006)	Golub train (38×3,051) + 8	RF backward elimination	RF vs. SVM, k-NN	.632+ bootstrap	error: SVM .014; RF .051 (no selection)	low
[2] (2010)	MILE stage I+II	standardised U133+2	all-pairs linear SVM	3×30-fold CV + prospective cohort	CV 92.2%, stage II 88.1% (18 classes)	low
[27] (2022)	Golub + 3	GNN-based ranking	SVM	10-fold CV	acc. 0.971 (2 genes)	likely high (selection before CV)
[28] (2022)	Golub + 2	ensemble wrapper (k-NN)	stacked DNN	k-fold CV	acc. 99.6%	unclear (selection order)
[29] (2022)	"Golub" (UCI)†	PSO (2,323 genes)	optimised CNN	random 70/30, 80/20	acc. 99.9%	high (no CV)
[30] (2024)	Golub + CuMiDa	RF importance ≥ 0.04	super learner	10-fold CV + 20% test	acc. 0.98	high (ranking before CV)

†The paper also lists CML, CLL and healthy classes, which the Golub data do not contain.

samples (27 ALL / 11 AML) and 34 independent samples (20 / 14), 65%/35% overall.

MILE (similar data set 1). Microarray Innovations in LEukemia, stage 1, GEO GSE13159 (DOIs 10.1200/JCO.2009.23.4732, 10.1111/j.1365-2141.2008.07261.x) [2], [5]. It contains 2,096 diagnostic samples in 18 classes, profiled with standardised protocols in 11 laboratories on Affymetrix HG-U133 Plus 2.0 (GPL570, 54,675 probe sets) from quantile-normalised signal [35]. We map the 14 ALL and AML classes to the binary label and exclude CLL, CML, MDS and non-leukemic samples. This leaves 1,292 samples (750 ALL / 542 AML; 1,211 bone marrow, 81 peripheral blood). *Why it was chosen:* same clinical question, a much larger multicentre cohort, Affymetrix technology, and a data set large enough for pre-training (bonus B1).

Andersson (similar data set 2, bonus B4). 121 consecutive childhood acute leukemias from Lund, Sweden, GEO GSE7186 (DOI 10.1038/sj.leu.2404688) [6]: 87 B-lineage ALL, 11 T-cell ALL and 23 AML (6 normal bone-marrow controls excluded). According to GEO, 108 samples are bone marrow and 13 peripheral blood. Platform: Swegene Human 27K RAP two-colour spotted cDNA array (GPL4861, 21,370 reporters). Values are lowess-normalised $\log_2$(sample/Universal Human Reference) ratios, centred per hospital and pre-filtered for 40% presence, so 7.19% of the values are missing. *Why it was chosen:* same question but a different technology, a fully paediatric series and real missing values.

Licences. NCBI places no restrictions on GEO data, although submitters may hold rights; all sets are used for academic research with citation.

Harmonisation. Probes were mapped to HGNC symbols with the GEO platform annotations (GPL80/GPL570/GPL4861). Ambiguous probes ("`A///B`") were removed and multiple probes per gene were collapsed to the probe with the highest mean expression. This gives 5,268, 20,848 and 9,677 genes, with 5,207 genes shared by Golub and MILE, 3,487 by Golub and Andersson, and 3,469 by all three; the platforms differ in scale, probe design and gene coverage (Section IV-F).

TABLE II
THE THREE DATA SETS SIDE BY SIDE.

	Golub	MILE	Andersson
Source	[1], Kaggle (CC0)	GSE13159	GSE7186
Platform	Hu6800 (GPL80)	U133 Plus 2.0 (GPL570)	27K cDNA (GPL4861)
Technology	1-colour oligo	1-colour oligo	2-colour cDNA
Measure	MAS4 avg. diff.	range-scaled log	$\log_2$ ratio
Population	mostly paediatric	multicentre	paediatric
Samples n	72	1,292	121
Features p	7,129	54,675	21,370
ALL / AML	47 / 25	750 / 542	98 / 23
Split	38 train / 34 test	none	none
Missing values	0	0.08%	7.19%
Genes (symbols)	5,268	20,848	9,677

IV. EXPLORATORY DATA ANALYSIS

All analyses were run in Python (pandas, SciPy, scikit-learn [36], seaborn) by `scripts/run_eda.py` with seed 42. *Note on leakage:* the supervised gene rankings in this section describe the data only. They must never be reused to select features for the models, because that would reproduce the selection bias of [19].

A. Data quality

Golub. The 513,288 values contain 0 missing or infinite entries, so the "fill INF with the mean" step of the module has no effect on the original files (it matters only after a log of non-positive values, which clipping avoids). However, 28.7% of the values are negative and 32.7% lie below 20, while 0.83% exceed 16,000 (range -28,400 to 71,369; Fig. 1a). Negative average differences arise when the mismatch signal exceeds the perfect-match signal, i.e. for transcripts that are absent. The calls confirm this: 70.2% of the measurements are Absent, 28.2% Present and 1.6% Marginal. The median

value is 33 for Absent and 662 for Present calls (Fig. 1c), and 1,971 probes are never called Present. The Present rate per array varies from 19.1% to 40.5%, with no visible class pattern (Fig. 1b). **Label mapping:** the patient columns are *not* in numeric order (training order 1–27, 34–38, 28–33). Labels were therefore joined on the patient number, and the script asserts that training IDs are exactly 1–38, test IDs 39–72 and that the class counts equal 27/11 and 20/14.

MILE and Andersson. Although GEO describes the MILE values as normalised signal, every array spans exactly 0.00–1.00. Comparing one array with its ArrayExpress sample table shows that the values are a monotone (Spearman 1.000) and almost linear function of $\log_2$ signal (Pearson 0.998 vs. 0.51 with the linear signal): they are per-array range-scaled log intensities, so no further log transform is applied. 45 probe sets are empty in every array and were dropped (0.08% of all values); there are no infinite entries. Andersson has 7.19% missing values: 12,084 reporters have at least one gap and 2,841 more than 20%; one array lacks 24.3% of its values (Appendix, Fig. 15). Imputation is therefore required, and it must be fitted inside the training folds.

B. Distributions and transformation

Values were clipped to the window $[20, 16000]$ given in the course guide and $\log_2$-transformed (Dudoit et al. [11] used a floor of 100, the same ceiling and base-10 logs). The pooled skewness falls from 7.08 to 0.55, and the Spearman correlation between gene mean and standard deviation drops from 0.95 to 0.18 (Appendix, Fig. 8). The transform thus removes most of the mean–variance dependence that would otherwise let a few high-intensity genes dominate distance-based methods (SVM, k-NN, PCA). After clipping, about one third of the values sit at the floor ($\log_2 20$), visible as a spike in the density. The filter of [11] (max/min $>$ 5 and max$-$min $>$ 500) keeps 3,859 of the 7,129 probes; with a floor of 100 it keeps 3,571, exactly the gene count of [11]. MILE (range-scaled log) and Andersson ($\log_2$ ratio, median -0.04) are already on log scales.

C. Dimensionality reduction

PCA on the centred, filtered $\log_2$ data (Fig. 2) shows that the variance is spread out: PC1 explains 10.8% and the first ten PCs 49.6%, and 37 PCs are needed for 80%. Even so, PC1 alone orders ALL and AML with an AUC of 0.984, i.e. the class difference is the largest single source of variation. t-SNE (perplexity 10) [37] shows two clear groups (silhouette 0.40 vs. 0.17 in 10-D PCA space), and the independent samples (black edges) mix with the training samples. In MILE (top 5,000 variable probes) the best single PC reaches AUC 0.987. In Andersson PC1 explains 17.5% with AUC 0.999.

D. Discriminative genes

For each gene we computed Welch's t-test, the Mann–Whitney U test (both with Benjamini–Hochberg FDR [38]) and Golub's signal-to-noise ratio (SNR). In Golub, 945 of 3,859 genes are significant at $q < 0.05$ (567 at $q < 0.01$; 1,014 with Mann–Whitney), 462 higher in AML and 483 higher in ALL. SNR and t rank genes almost identically ($\rho = 0.999$; Appendix C). The top genes (Appendix, Table IV) agree with known biology: myeloid genes (CST3, CD33, MPO, ZYX, MGST1) are higher in AML, and lymphoid/B-cell genes (TCF3, CD79B, CD19, DNTT, VPREB1, CCND3) are higher in ALL. CST3, ZYX and CD33 are also among the 50 informative genes of [1]. Hierarchical clustering of the samples on the top-30 genes (Fig. 3) recovers the classes (adjusted Rand index ARI = 0.89). In Andersson the ARI is only 0.41: the T-ALL samples cluster next to AML, because they lack the B-cell programme that drives most ALL-high genes. With 1,292 samples, 69.1% of the MILE probe sets are significant, clustering again recovers the classes (ARI 0.86), and LEF1 (lymphoid) and MGST1, CFD, APLP2 (myeloid) lead the ranking.

E. Training vs. independent set (batch effect)

The mean Present rate is 29.0% in the training set and 27.3% in the independent set (Mann–Whitney $p = 0.071$). The 75th-percentile intensity, a proxy for array brightness, is lower in the independent set (median 421 vs. 482, $p = 0.016$; Appendix, Fig. 6). To separate batch from class composition we compared train and test *within* each class with per-gene Kolmogorov–Smirnov tests. Under no shift, 5% of the genes would have $p < 0.05$. We observe 17.7% for ALL (28 genes significant after FDR) and 8.0% for AML (0 genes). So there is a moderate, ALL-specific shift. This is consistent with the more heterogeneous collection of the independent samples and may be one reason why a single 34-sample test score can differ from CV estimates (**RQ2**), although [11] saw no clear loss of accuracy on this split.

F. Cross-cohort comparison

On the common genes, the gene-wise t statistics are positively correlated between cohorts (Fig. 4a,b): $\rho = 0.56$ for Golub–MILE, $\rho = 0.44$ for Golub–Andersson and $\rho = 0.67$ for MILE–Andersson. Of the top 100 genes, 33 are shared by Golub and MILE (hypergeometric $p = 3.6 \times 10^{-28}$; 100% with the same direction) and 28 by Golub and Andersson ($p = 1.5 \times 10^{-21}$). 18 genes are in the top 100 of all three cohorts (CST3, CD33, CTSA, POU2AF1, CD79A, PYGL, PCBD1, NPY, ASAH1, MAPKAPK3, CD19, CSTA). The *scales*, however, are incompatible: median expression is 7.31, 0.46 and -0.03 on the three log scales. In a pooled PCA without harmonisation, cohort explains $\eta^2 = 99.8\%$ of PC1 and class only 0.0% (Fig. 4c). After per-cohort gene-wise z-scoring, cohort explains 0.0% and class 61.4% (Fig. 4d). The biology transfers across cohorts but the measurement scale does not, so cross-cohort models will use per-cohort standardisation or ComBat [24], estimated without test labels; because class proportions differ between cohorts, these adjustments can remove class signal, which will be checked.

G. Numbered findings and implications for modelling

1) *Negative and floor-level intensities* (28.7% < 0) $\Rightarrow$ clip to $[20, 16000]$ and apply $\log_2$ before any scale-sensitive model.

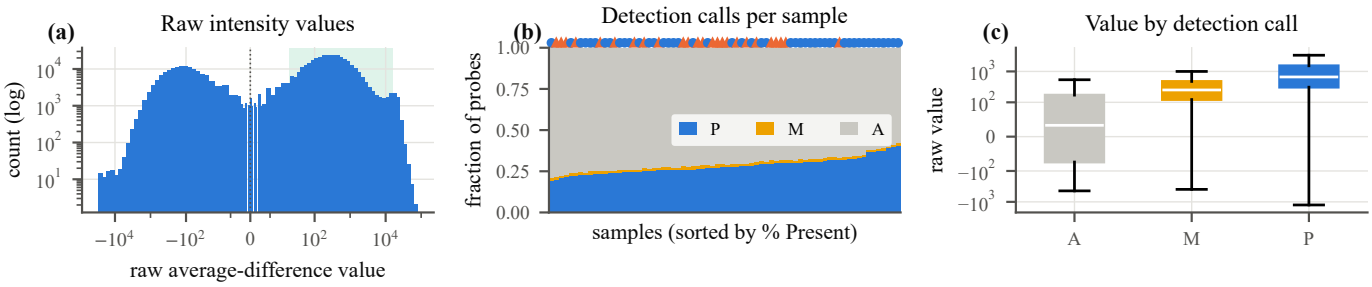


Fig. 1. Golub data quality. (a) Raw average-difference values on a symmetric-log axis; the shaded band is the clipping window [20, 16000]. (b) Fraction of Present/Marginal/Absent calls per array (markers on top: class). (c) Raw value by detection call: Absent calls coincide with negative or near-zero values.

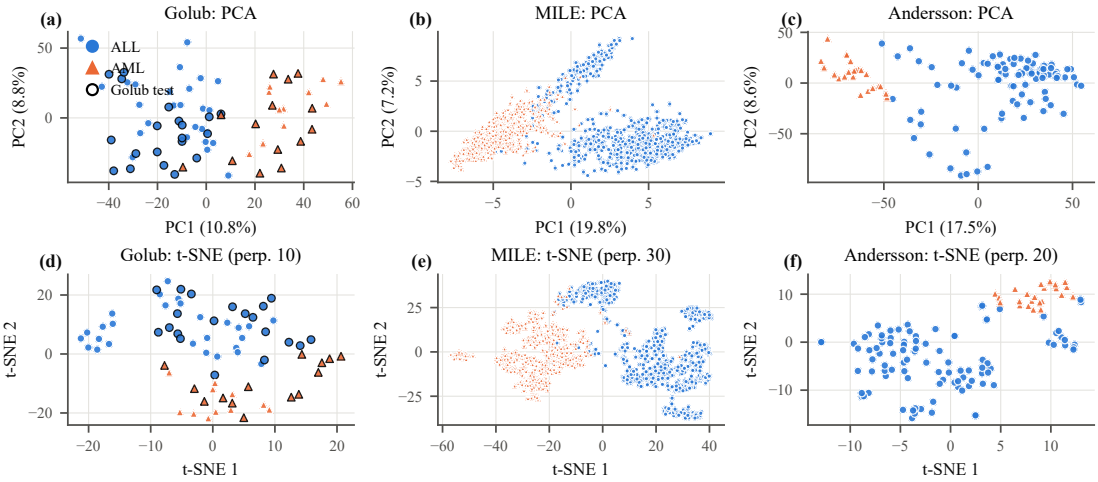


Fig. 2. Unsupervised embeddings of the three cohorts (top: PCA, bottom: t-SNE; colour/marker: class). Golub independent-set samples have black edges. All cohorts show a dominant ALL/AML axis; scree plots are in Appendix C.

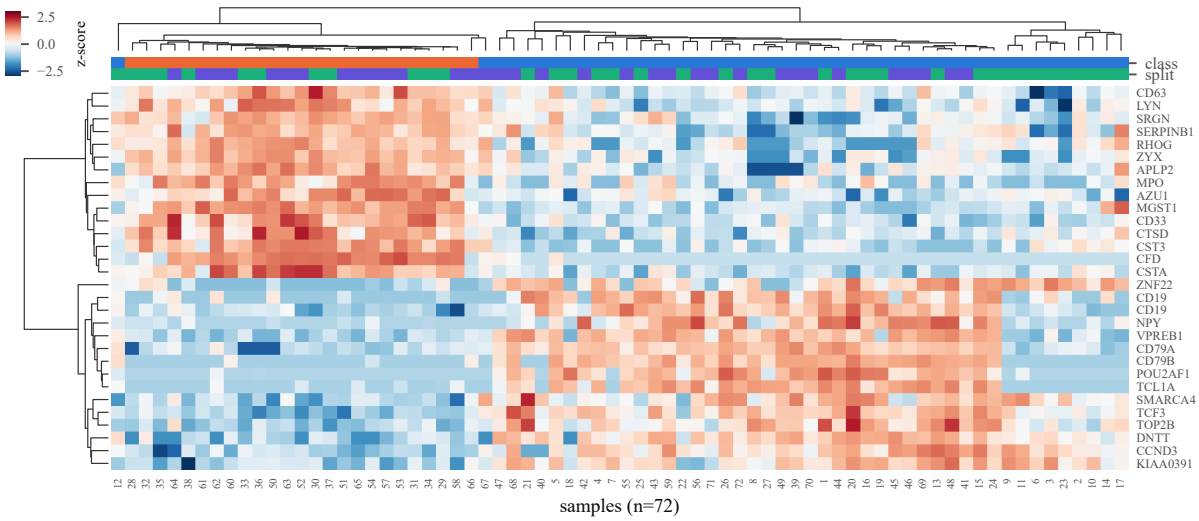


Fig. 3. Golub: z-scored expression of the 15 most AML-high and 15 most ALL-high genes (Welch t), average linkage, $1-$Pearson distance. Column bars: class (blue ALL, orange AML) and split (green train, violet test). Floor-clipped values appear as uniform light-blue blocks.

2) $p \gg n$ (7,129 features, 72 samples) $\Rightarrow$ feature selection (variance or fold-change filter, then univariate k-best) *inside* each training fold, with k tuned in the inner CV. The ablation will cover $k \in \{10, \ldots, 500\}$ and PCA.

3) *Strong low-dimensional signal* (PC1 AUC 0.984) $\Rightarrow$ near-perfect scores are plausible; metric differences between methods will be small, so confidence intervals and paired tests are essential (**RQ1**).

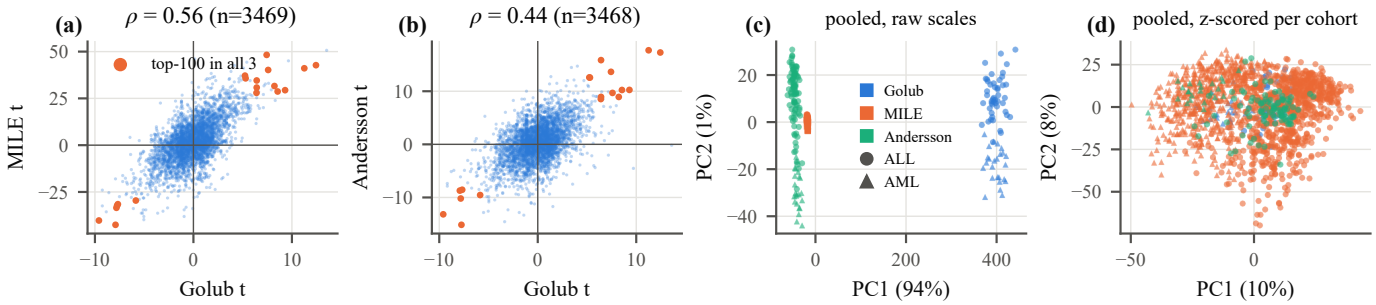


Fig. 4. Cross-cohort compatibility on shared genes. (a,b) Gene-wise Welch t statistics of Golub vs. MILE and vs. Andersson (orange: genes in the top-100 of all three). (c) Pooled PCA without harmonisation separates cohorts, not classes; (d) per-cohort z-scoring brings out the ALL/AML axis (colour: cohort, marker: class).

4) *Class imbalance* (65/35, MILE 750/542, Andersson 98/23) ⇒ stratified folds, `class_weight=balanced`, and balanced accuracy, F1 and ROC-AUC alongside accuracy. SMOTE only within training folds, if used at all.
5) *Train/test shift* in Golub ⇒ report the 38/34 result and nested CV separately and explain the difference (**RQ2**).
6) *Missing values* (Andersson) and *scale sensitivity* ⇒ in-fold median/k-NN imputation and in-fold z-scoring for LR, SVM and neural nets (trees are scale-invariant).
7) *Shared biology but different scales* ($\rho > 0$, significant overlap, PC1 dominated by cohort) ⇒ per-cohort standardisation on the 3,469 common genes for cross-cohort transfer (**RQ3**). Lineage heterogeneity (T-ALL) is a likely source of errors and will be analysed.

V. Plan for the Final Report

Methods (no results yet). Three ML methods, chosen to cover different inductive biases: (M1) *logistic regression with the elastic-net penalty* [16], a sparse linear model with interpretable coefficients; (M2) an *SVM* with linear and RBF kernels, the historical reference for microarrays [9], [10]; and (M3) a *random forest*, which models interactions, is invariant to monotone transformations [13] and offers permutation importance [14]. The module's decision tree is kept as a baseline.

Pipeline and validation. Every data-dependent step of the pipeline (Appendix B, Fig. 5) is a scikit-learn transformer (`src/leukemia/preprocessing.py`) re-fitted inside each fold. Two protocols: (a) the module's 38/34 split, tuned by 5-fold stratified CV on the 38 samples and evaluated *once* on the 34 after model selection; (b) nested CV on all 72 samples, outer 5-fold stratified CV repeated with 5 seeds (42, 7, 123, 2024, 31337), inner 5-fold search, identical outer folds for all methods (paired design). MILE and Andersson use the same protocol; cross-cohort transfer uses the common genes with per-cohort standardisation. **Search spaces:** EN-LR $C \in [10^{-3}, 10^{2}]$ (log), ℓ_1 ratio $\in \{0.1, 0.5, 0.9, 1\}$; SVM $C \in [10^{-3}, 10^{3}]$ (log), $\gamma \in [10^{-5}, 10^{-1}]$; RF #trees $\in \{300, 1000\}$, max-features $\in \{\sqrt{p}, 0.1p, 0.3p\}$, min-leaf $\in \{1, 2, 4\}$; $k \in \{10, \ldots, 500\}$ for all. Grid search (EN-LR, SVM), randomised search (RF) and Optuna TPE (neural nets) will be compared by cost and inner-CV score.

Metrics and statistics. Accuracy, balanced accuracy, precision, recall (sensitivity), specificity, F1, ROC-AUC and confusion matrices, as mean ± SD over 25 outer folds with approximate 95% CIs from the corrected resampled t [3]; pairwise tests with Holm adjustment within a cohort, descriptive average ranks across the three cohorts [22], and McNemar on the fixed split (two trained models only); ROC/PR, loss and k–performance curves.

Bonuses. *B1:* MLP (dropout, weight decay) and autoencoder + classifier, with early stopping and architecture choice in the inner folds and optional pre-training on MILE. *B2:* TabPFN [33] (pinned version) after in-fold selection of ≤ 500 genes, the upper end of its evaluated range, and GenePT-style embeddings [34], an untested adaptation to bulk microarrays. *B3:* an English paper for an international IEEE-sponsored conference open in the final week (with the instructor's approval). *B4:* Andersson as the third data set. **Timeline:** weeks 1–2 Golub nested CV (**RQ1**,**RQ2**); 3–4 MILE/Andersson, transfer and ablations (**RQ3**); 5–6 B1/B2 (**RQ4**) and interpretability; 7 paper, video, B3. **Risks:** variance at $n = 72$ (repeated nested CV, CIs); batch/platform effects (per-cohort scaling, ComBat check); neural overfitting (early stopping); TabPFN's validated range (in-fold selection); disease information leaking through gene text summaries (B2).

VI. Reproducibility and Conclusion

`python run_all.py` rebuilds this PDF from raw data (seed 42, pinned `requirements.txt`); every number in the text is a LaTeX macro written by the code. In summary, all three cohorts carry a strong, biologically coherent ALL/AML signal; the obstacles to valid modelling are $p \gg n$, the Golub train/test shift and cross-platform scale differences. The final report will test **RQ1**–**RQ4** with the protocol above.

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Appendix A
Full literature table

TABLE III: Full literature summary table (all 30 included studies, chronological).

Ref.	Year	Data set	Pre-proc./feature sel.	Method	Validation	Best result	Note
[1]	1999	Golub ALL/AML (Affymetrix Hu6800)	Informative-gene selection by class correlation	Class discovery + weighted-vote class predictor	LOOCV on 38 plus fixed 34	LOOCV 36/38 called; test 29/34 called, all correct; 5 uncertain	Original dataset; mostly paediatric
[9]	2000	Ovarian, Golub, colon	n/r	SVM	Hold-one-out plus 38/34	Golub test 30-32/34 correct	No clear difference between techniques
[4]	2002	MLL/ALL/AML leukemias	n/r	Expression profiling of MLL-rearranged leukemia	n/r	n/r	Common 3-class leukemia benchmark
[10]	2002	Leukemia and colon microarrays	SVM-RFE gene ranking	Linear SVM	LOO and 38/34	Golub test 0/34 errors with 8 or 16 genes	Selection not stated to be inside LOO
[11]	2002	Golub, NCI60, lymphoma	Floor 100, ceiling 16000, filter, log10	LDA/DLDA, kNN, trees, aggregation	150 repeated 2:1 splits	DLDA median error 0 (two-class leukemia)	Technical report 576 read, not the JASA version
[12]	2002	SRBCT and leukemia	Shrunken centroid soft-thresholding	Nearest shrunken centroids (PAM)	10-fold CV on 38 plus fixed 34	CV 1/38, test 2/34, 21 genes	Nearest shrunken centroids
[19]	2002	Colon and Golub	Gene selection inside vs outside CV	SVM-RFE; Fisher with forward selection	External 10-fold CV; .632+ bootstrap	Selection bias about 5% (38/34), 2-3% (all 72)	Selection bias; CV error not zero
[15]	2005	11 datasets incl. leukemia	Several gene selection methods	Multicategory SVMs vs kNN/NNs/ensembles	Two cross-validation designs	n/r	MC-SVMs best; GEMS system
[16]	2005	Methodology (incl. microarray example)	Elastic net penalty	Penalized regression	n/r	10-fold CV 3/38, test 0/34, 45 genes	Squared loss on 0-1 response
[13]	2006	Golub train 38 x 3,051 + 8 other sets	RF backward gene elimination	Random forest vs SVM/kNN/DLDA/SC	.632+ bootstrap	SVM .014; RF .051 without, .075-.087 with selection	Leukemia RF selects 2 genes
[20]	2006	Simulation only (40 samples, 6,000 features)	Tuning within CV	Nested CV	Nested vs non-nested CV	Nested estimates close to true error	Almost unbiased, slightly pessimistic
[6]	2007	121 childhood leukemias (GSE7186)	n/r	k-nearest neighbor predictors	Independent data set validation	T-ALL MRD status classified with 100% accuracy	External validation source (pediatric)
[24]	2007	Two example data sets	Empirical Bayes location/scale adjustment	ComBat	n/a	n/r	Name ComBat not in the paper
[5]	2008	MILE pre-phase (multi-lab)	Standardized protocols across labs	Inter-laboratory reproducibility study	Cross-lab comparison	n/r	Linked to GSE13159 in GEO
[2]	2010	MILE stage I/II (GSE13159 = stage I)	Whole-genome HG-U133 Plus 2.0 profiling	All-pairs linear SVM	Stage I CV; stage II prospective, custom chip	Stage I 3x30-fold CV 92.2%; stage II 88.1%	External validation source; 11 labs
[21]	2010	13 benchmark sets plus one synthetic	Hyperparameter selection	KRR, KLR, GPC	Model-selection overfitting analysis	n/r	Selection variance matters; no DOI
[23]	2010	Review plus reanalysis of nine data sets	Batch effect diagnostics	Review	n/a	n/r	Motivates cross-cohort batch handling
[7]	2019	CuMiDa (78 curated GEO sets)	Background correction + normalization + QC	Benchmark ML baselines	n/r	n/r	Curated microarray benchmark repository
[25]	2020	12,029 samples from 105 studies	High-dimensional genome-wide input	ML for AML prediction	100 random 80/20 splits; study-disjoint; cross-platform	n/r	AML vs all others including ALL
[26]	2021	>16,400 blood transcriptomes (incl. leukaemia)	n/r	Swarm Learning (decentralized ML)	Simulated silos of pooled data; global test node	n/r	Central model slightly better on AML
[32]	2021	11 data sets, n 20,640 to 1.2M	n/r	FT-Transformer / ResNet vs GBDT	Benchmark protocol	n/r	Best deep model on most; GBDT wins some
[31]	2022	45 data sets, n>=3,000, d<500	n/r	Tree models vs deep learning	Benchmark protocol	n/r	Regime excludes p»n microarrays
[17]	2022	132 reviewed studies	Feature selection methods	Systematic review	n/a	n/r	Most frequent research direction (34.9%)
[27]	2022	Golub leukemia (72 x 7129) + 3 others	GNN-based feature ranking aggregation	SVM classifier	10-fold CV	Leukemia Acc 0.971 with 2 genes	Selection described before 10-fold CV
[28]	2022	Golub leukemia (72 x 7129) + DLBCL + prostate	Soft-ensemble wrapper gene selection (kNN)	Stacked deep neural network	k-fold CV (K=5-20)	Leukaemia average accuracy 99.6%	Gene selection vs CV order unclear
[29]	2022	Golub leukemia (72 x 7129; UCI)	PSO selected 2323 of 7129 genes	HORD-optimized CNN	Random 70/30 and 80/20 splits	Leukemia accuracy 99.9% (comparison table)	No CV; CML, CLL, healthy classes reported, not in Golub
[18]	2023	Microarray and RNA-seq (review)	Feature engineering + preprocessing	Review of ML/DL classifiers	n/a	n/r	Covers MLP/CNN/RNN/GNN/transformers
[30]	2024	Golub ALL/AML + CuMiDa leukemia	RF feature importance (FI >= 0.04)	Super learner (RF metalearner)	Stratified 10-fold CV with 20% test	AML-ALL accuracy 0.98 (Table 2)	FI ranking appears done before CV
[33]	2025	Up to 10,000 samples, 500 features, 10 classes	n/r	TabPFN foundation model	Benchmark comparisons	n/r	n around 72 untested
[34]	2025	Gene-level and scRNA-seq tasks	NCBI summaries, text-embedding-ada-002	GenePT embedding model	Downstream task benchmarks	n/r	Preprint read, not journal version

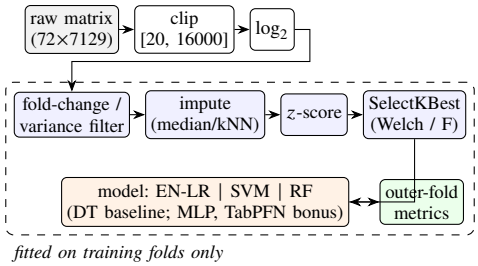


Fig. 5. Planned leakage-free pipeline. Stateless steps (clip, $\log_2$) may precede the split; every step that learns from data (dashed box) is an `sklearn Pipeline` stage re-fitted in each inner and outer training fold.

Appendix B
Planned pipeline

Appendix C
Additional EDA figures

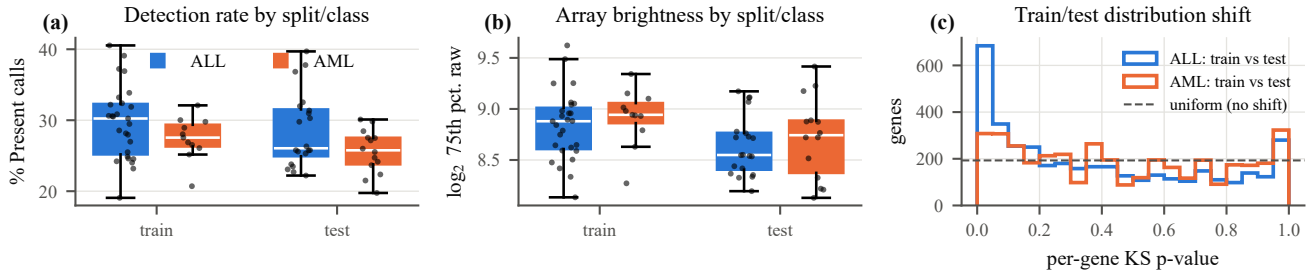


Fig. 6. Golub train vs. independent set. (a) Present-call rate and (b) array brightness by split and class. (c) Per-gene KS p-values comparing train and test within each class; the excess near 0 for ALL indicates a distribution shift.

TABLE IV
Top 12 genes by Welch p in each cohort (Δ: mean difference AML−ALL on the cohort's log scale; $t > 0$: higher in AML).

	Golub			MILE			Andersson		
#	gene	Δ	t	gene	Δ	t	gene	Δ	t
1	MGST1	+4.10	+13.5	PNKD	+0.25	+56.8	PILRB	-2.95	-16.0
2	CST3	+5.82	+12.4	LEF1-AS1	-0.35	-59.5	INOC1	-1.61	-15.0
3	TCF3	-2.07	-11.8	LEF1	-0.31	-52.5	NPY	-5.62	-15.1
4	ZYX	+3.83	+11.0	CFD	+0.43	+49.8	–	–	–
5	CD33	+2.26	+11.3	APLP2	+0.21	+49.3	NPR1	-3.31	-14.4
6	APLP2	+2.77	+9.6	LOC101928615	+0.22	+48.8	PCBD1	+2.98	+15.9
7	CD79B	-4.45	-9.2	MGST1	+0.38	+53.2	C17orf58	+2.11	+16.7
8	POU2AF1	-3.09	-9.6	MGST1	+0.41	+50.7	SPG20	+3.07	+13.9
9	DNTT	-4.44	-9.4	LEF1	-0.37	-55.1	LOC201229	-3.50	-14.7
10	CCND3	-2.06	-9.2	CTSA	+0.20	+48.2	CTSC	+2.38	+14.8
11	VPREB1	-4.24	-8.6	FUCA2	+0.24	+47.9	PCBD1	+2.99	+14.6
12	RHOG	+3.17	+8.6	LEF1-AS1	-0.25	-47.3	CD33	+3.87	+17.7

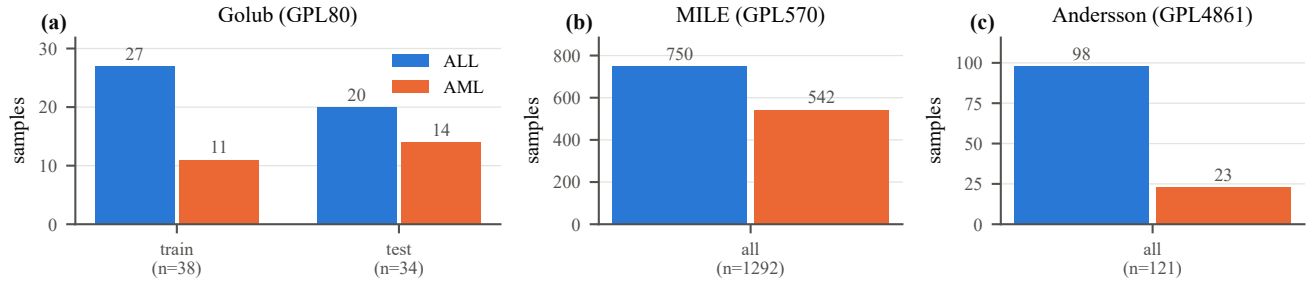


Fig. 7. Class distribution. (a) Golub reference split; (b) MILE after mapping 14 ALL/AML subclasses to two labels; (c) Andersson after removing normal controls. All sets are imbalanced towards ALL.

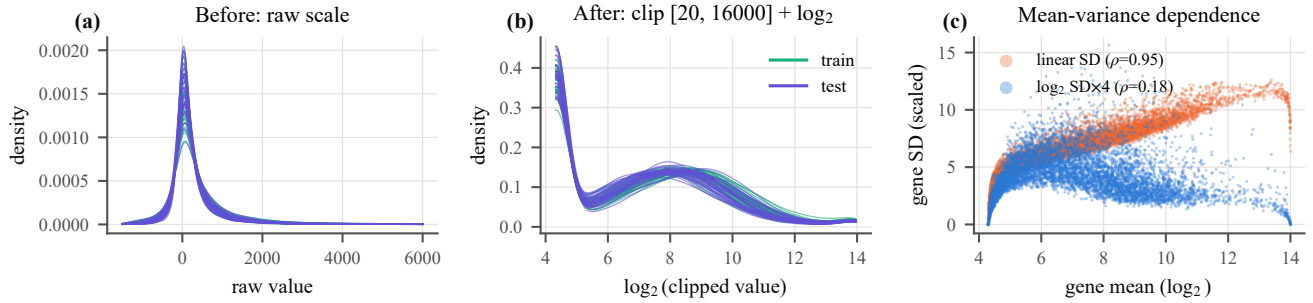


Fig. 8. Golub per-array densities (a) before and (b) after clipping and $\log_2$ (colour: reference split). (c) Gene standard deviation vs. mean: on the linear scale SD grows with the mean ($\rho = 0.95$), after $\log_2$ the dependence largely disappears ($\rho = 0.18$).

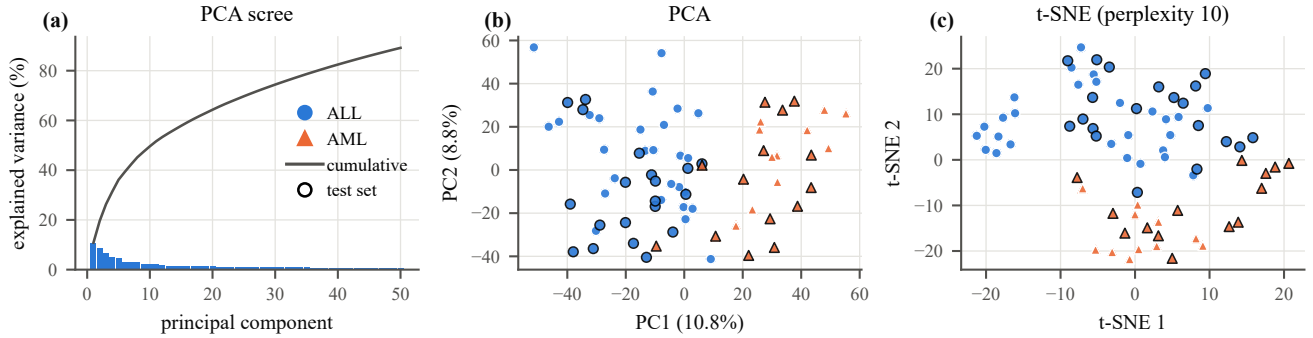


Fig. 9. Golub: PCA scree plot (bars: per component, line: cumulative), PCA and t-SNE. PC1 explains 10.8% and ten PCs 49.6%; 37 PCs are needed for 80%. The variance is spread over many directions, yet PC1 alone separates the classes.

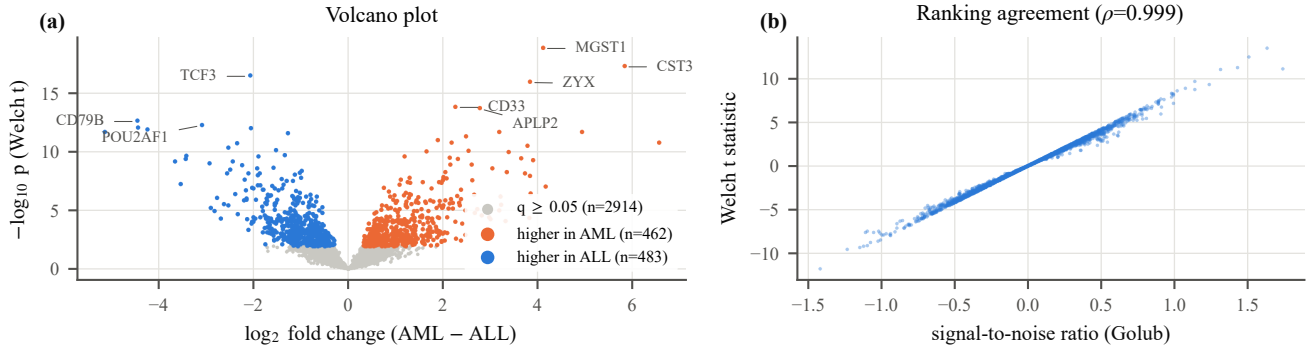


Fig. 10. Golub: (a) volcano plot of Welch tests (colour: BH $q < 0.05$): 462 genes are higher in AML and 483 in ALL, i.e. the differences are not one-sided. (b) Golub's signal-to-noise ratio and Welch t rank genes almost identically ($\rho = 0.999$), so the choice between them matters little for a univariate filter.

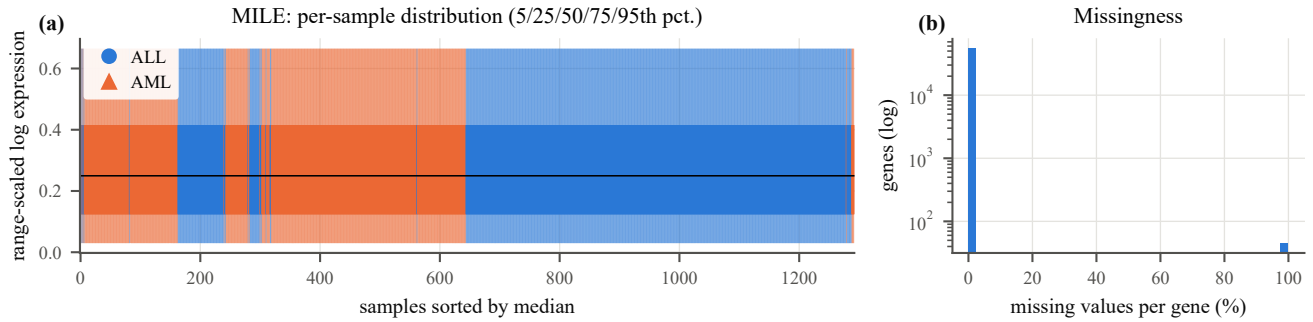


Fig. 11. MILE: (a) per-array quantiles of the range-scaled log values (sorted by median). All arrays have identical quantiles, as expected after quantile normalisation, so there are no array-level outliers. (b) Missing values per probe set: 45 probe sets are empty in every array (removed); all others are complete.

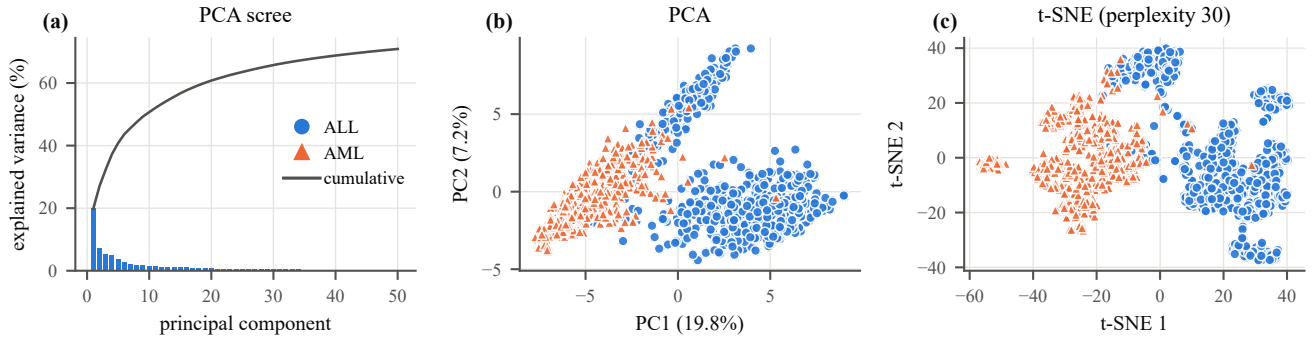


Fig. 12. MILE: PCA scree, PCA and t-SNE (top 5,000 most variable probes). PC1 explains 19.8% and separates ALL from AML (AUC 0.987); in t-SNE the ALL samples form several sub-clusters, in line with the many ALL subtypes in MILE.

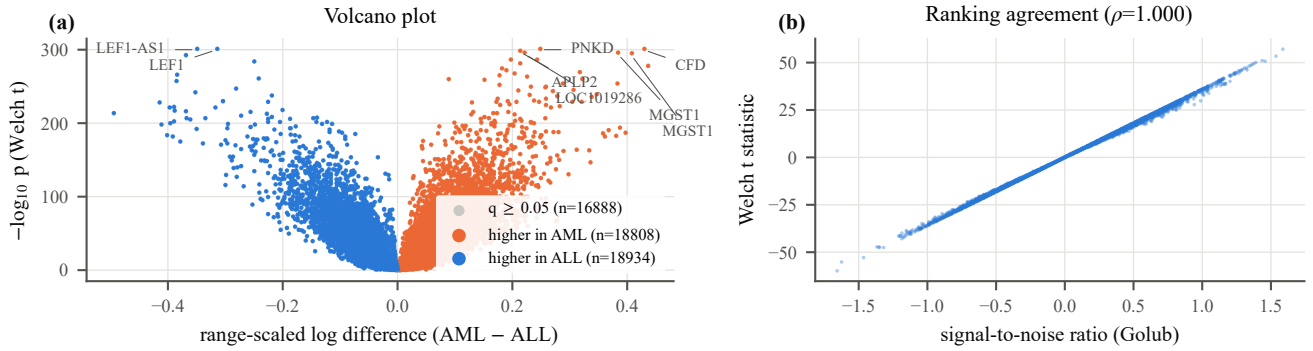


Fig. 13. MILE: (a) volcano plot; the x axis is a difference of range-scaled log values, and $p < 10^{-300}$ is drawn at 300. With $n = 1{,}292$, 69.1% of probe sets reach $q < 0.05$, so effect size rather than p should guide gene selection. (b) SNR–t agreement.

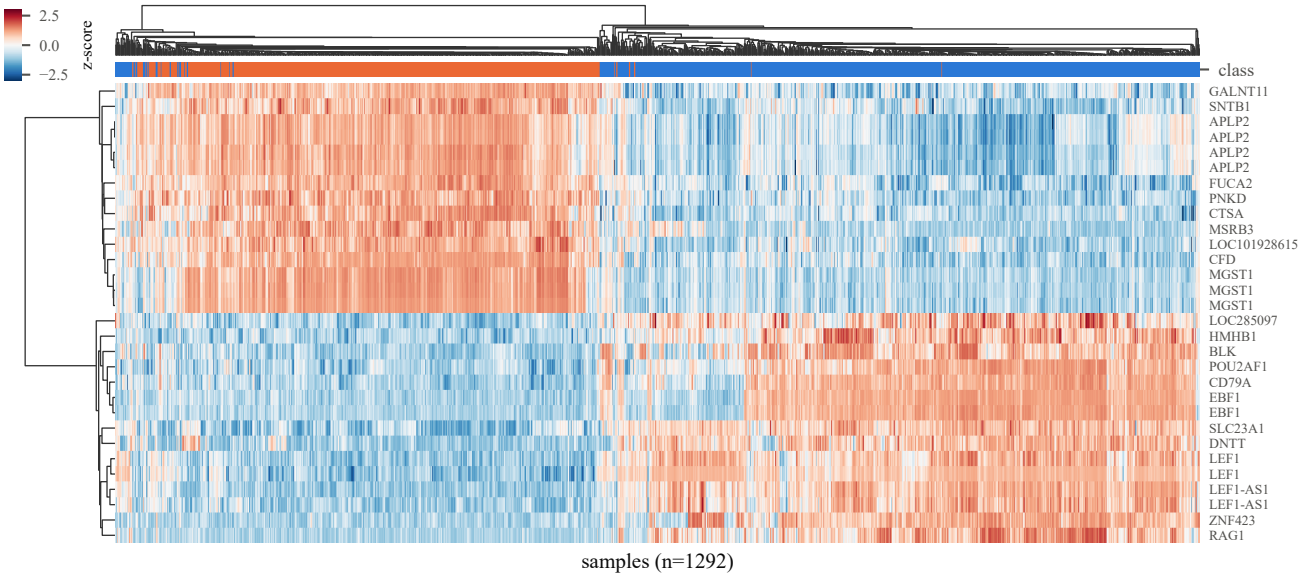


Fig. 14. MILE: clustered heatmap of the top-30 genes. The two top-level clusters match ALL/AML (ARI 0.86); within ALL, a sub-cluster lacks the B-cell genes (EBF1, CD79A, BLK, POU2AF1), consistent with T-ALL.

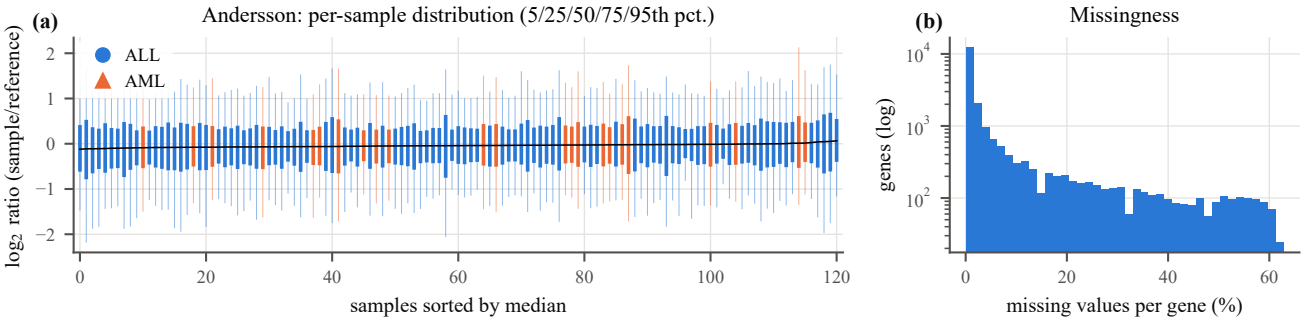


Fig. 15. Andersson: (a) per-array $\log_2$-ratio distributions are centred near 0 with similar spread, as expected after lowess normalisation and centring; (b) missing values per reporter: 12,084 reporters have gaps and 2,841 miss more than 20%, so imputation (in-fold) is required.

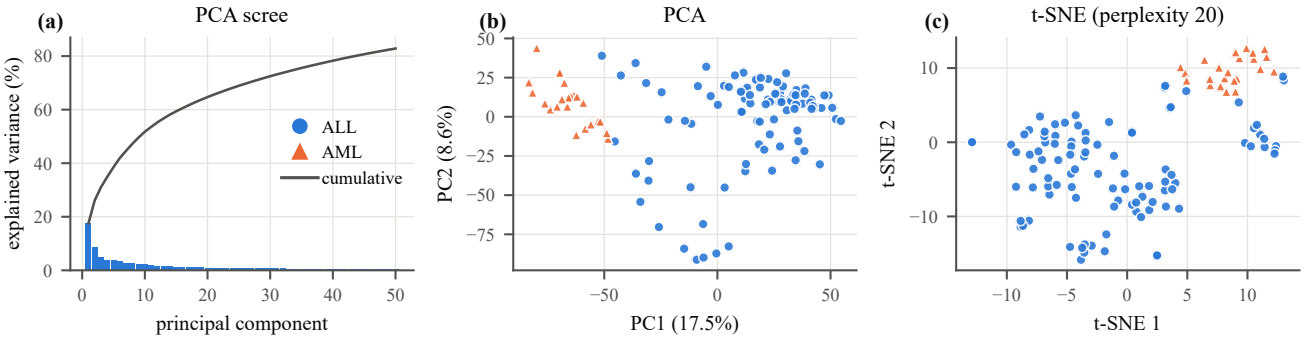


Fig. 16. Andersson: PCA scree, PCA and t-SNE (median-imputed for visualisation only). PC1 explains 17.5% and almost perfectly orders AML vs. ALL (AUC 0.999); t-SNE shows a separate AML group and several ALL sub-groups.

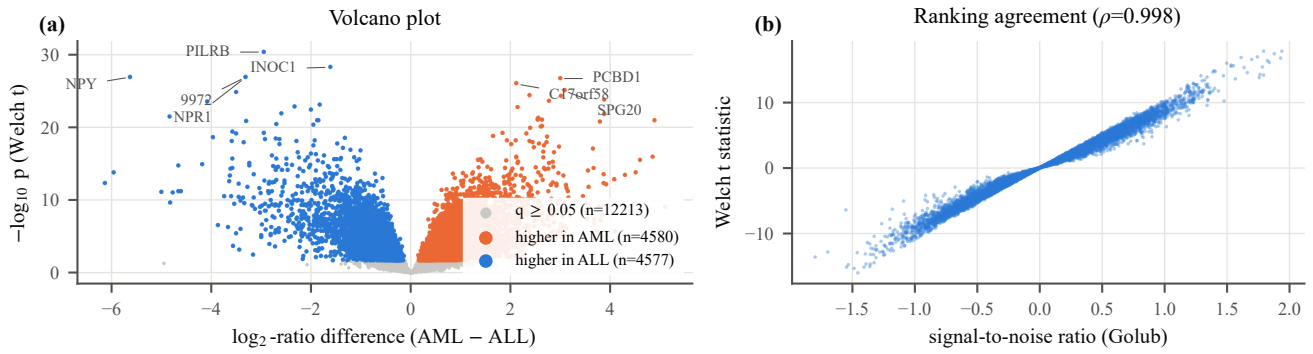


Fig. 17. Andersson: (a) volcano plot (x: difference of $\log_2$ ratios); 42.8% of reporters reach $q < 0.05$, split evenly between AML-high (4,580) and ALL-high (4,577). (b) SNR–t agreement.

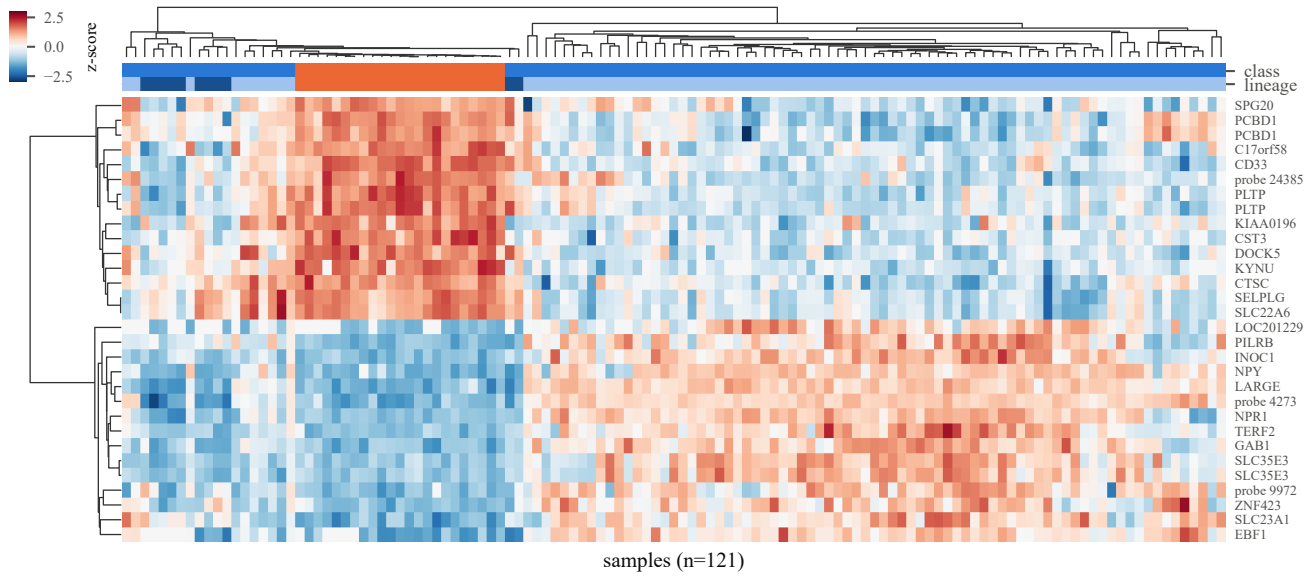


Fig. 18. Andersson: clustered heatmap of the top-30 genes. Lineage bar: light blue B-lineage ALL, dark blue T-cell ALL, orange AML. T-ALL samples cluster next to AML.